

Final Report: Newton–Schulz Schedule Design in Muon for Transformer Pretraining

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1 Problem Setup

Problem statement Muon is a recently proposed optimizer that updates matrix parameters by orthogonalizing a momentum-like update direction through a short Newton–Schulz (NS) iteration [2, 3]. Recent theory has also started to analyze the convergence properties of Muon-style NS updates [4]. In practice, the iteration is controlled by quintic coefficients, so schedule design becomes a spectral design problem rather than a minor implementation detail. This project asks: *under a fixed transformer pretraining stack, how do different NS coefficient schedules change the spectral geometry of Muon updates, and how does this spectral geometric change affect validation loss and training cost?*

Mathematical formulation To understand Muon, it is helpful to start from the limitation of AdamW. For a weight matrix $W \in \mathbb{R}^{m \times n}$, AdamW treats the gradient elementwise. This is effective, but it ignores the fact that W is a matrix with coupled row and column geometry. In particular, an update can have very uneven singular values, meaning that some directions are stretched much more strongly than others. Muon is motivated by the idea that for hidden-layer matrix parameters, a better update should acts more isotropically on the singular directions and avoids concentrating into a few dominant directions.

For a rectangular matrix X , X is semi-orthogonal if:

$$X^\top X \approx I_n \quad \text{if } m \geq n, \quad XX^\top \approx I_m \quad \text{if } m < n. \quad (1)$$

Equivalently, if $X = U\Sigma V^\top$ is an SVD, then the ideal semi-orthogonal factor is the polar factor $UV^\top$, whose nonzero singular values are all equal to 1. Muon utilizes an iterative algorithm called NS iteration instead of SVD to approximately orthogonalize the update matrix before applying it to the parameters.

Why can NS iterations produce a semi-orthogonal matrix? The reason is that they act on singular values. After a normalization step, Muon repeatedly applies a quintic matrix polynomial. If X_k is the matrix after the k -th NS iteration, then

$$X_{k+1} = a_k X_k + b_k (X_k X_k^\top) X_k + c_k (X_k X_k^\top)^2 X_k, \quad (2)$$

or the aspect-ratio-swapped equivalent when the matrix is tall. Because this map is equivariant to the SVD, it leaves singular vectors unchanged and transforms only the singular values. At the scalar level,

$$\sigma_{k+1} = p_k(\sigma_k), \quad p_k(\sigma) = a_k \sigma + b_k \sigma^3 + c_k \sigma^5. \quad (3)$$

Therefore, a coefficient schedule $\{(a_k, b_k, c_k)\}_{k=1}^T$ defines a composed spectral map on singular values. If the composed map pushes the relevant singular-value range toward 1, then the resulting matrix becomes closer to the polar factor and hence closer to semi-orthogonality. This is the mathematical core of Muon: orthogonalization is implemented indirectly by shaping the singular-value dynamics with a short polynomial iteration. The high level of a muon update procedure can be summarized as in Algorithm 1.

Algorithm 1 Muon update procedure

- 1: Initialize momentum state M_t
 - 2: **for** $t = 1, 2, \dots$ **do**
 - 3: Compute matrix gradient $G_t = \nabla_W \mathcal{L}_t(W_{t-1})$
 - 4: Update accumulated momentum $M_t \leftarrow \mu M_{t-1} + G_t$
 - 5: Apply NS iteration on M_t to obtain O_t
 - 6: Update parameters: $W_t \leftarrow W_{t-1} - \eta_t O_t$
 - 7: **end for**
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This perspective also explains why coefficient choice matters. Different coefficients create different scalar maps p_k : some are conservative and stable, some are more aggressive and rapidly amplify small singular directions, and some are designed adaptively from a presumed lower bound on the spectrum [1]. In other words, Muon is not just one optimizer; it is a family of optimizers parameterized by spectral maps. In this project, we systematically compare four coefficient schedules of Muon along with a AdamW baseline:

- **AdamW**: no NS orthogonalization on matrix parameters.
- **Stable**: conservative mapping coefficient $(2.0, -1.5, 0.5)$.
- **Manual**: fast-to-stable schedules
- **Fast**: aggressive mapping coefficient $(3.4445, -4.7750, 2.0315)$.
- **Polar Express (PE)**: dynamically compute coefficient at each step [1].

2 Technical Approach

Experimental design The philosophy of the experiment is to isolate the effect of the NS coefficient schedule itself. For that reason, we use a controlled design in which the architecture, data source, token budget, learning-rate schedule, batch structure, and all Muon hyperparameters other than the orthogonalization schedule are held fixed across runs. Under this design, observed differences can be interpreted as consequences of spectral-map design rather than of unrelated tuning changes.

AdamW is included as an external baseline because it provides a useful reference point for what happens when matrix parameters are not orthogonalized. Within the Muon family, we fix the NS iteration depth to $T = 5$ steps for the core schedule comparison. We further conduct several ablation studies on important Muon hyperparameters, especially the iteration depth and the PE lower-bound parameter. Finally, we combine the training results and mathematical analysis of the NS iteration to better interpret the relation between semi-orthogonal updates and validation loss in language pretraining.

Evaluation dimensions The study evaluates each method along three dimensions. First, we measure *optimization quality* through validation-loss trajectories, final validation loss, and token-normalized area under the validation-loss curve (AUC). Second, we measure *computational cost* through end-to-end wall-clock time, because a schedule that improves geometry but is too expensive would be less practically useful. Third, we measure *spectral geometry* directly by analyzing update matrices during training. This geometric analysis is essential: without it, any difference in final loss would remain a black-box empirical observation rather than a mathematically interpretable effect of the NS map.

Experimental setup All experiments are run on one RTX 4090 GPU in BF16 precision using FineWeb-10B tokenized shards [5]. The model is a standard 11-layer pre-norm Transformer with model dimension 768, 6 attention heads, head dimension 128, MLP ratio 4, and sequence length 2048. The total training budget is fixed at 100M tokens, with 131,072 tokens per optimizer step, 16 gradient-accumulation steps, 2% warmup, and cosine decay to 20% of the peak learning rate.

The optimizer settings are chosen to match a realistic small-transformer pretraining regime:

- AdamW baseline: learning rate 6×10^{-4} , $(\beta_1, \beta_2) = (0.9, 0.95)$, weight decay 0.1.
- Muon family: base learning rate 0.023, momentum 0.95, $\beta_2 = 0.9$, weight decay 1.2.

These values lie in the empirically established range commonly used in modern small-transformer pretraining practice and in public Muon baselines [3], so they should be viewed as reasonable default settings rather than arbitrary picks.

Muon variant used in this project The project does not use the simplest momentum SGD shown in Algorithm 1. The actual matrix update path is a *Nesterov-style* lookahead variant together with second-moment rescaling on the , so more precisely it is a normalized Muon implementation. Let G_t denote the matrix gradient at step t , and let B_t be the momentum buffer. The update path is

$$B_t = mB_{t-1} + (1 - m)G_t, \quad (4)$$

$$g_{\text{pre},t} = mB_t + (1 - m)G_t, \quad (5)$$

$$g_{\text{post},t} = \mathcal{P}(g_{\text{pre},t}), \quad (6)$$

where $\mathcal{P}$ is the schedule-defined NS orthogonalization map. The use of $g_{\text{pre},t}$ means the optimizer does not orthogonalize the raw buffer itself, but a Nesterov-style lookahead combination that effectively looks one step ahead along the momentum direction.

After orthogonalization, the implementation further applies a second-moment normalization on the matrix update magnitude before weight decay and the learning-rate step. In that sense, our project studies coefficient schedules inside the specific normalized Muon variant actually used by this training stack. This detail is important for interpreting the results, because the measured update geometry is the geometry of the real implemented search direction rather than of an idealized Muon formula.

For the spectral geometric study, every 10M training tokens we sample representative update matrices and record singular-value statistics together with a short-side semi-orthogonality error:

$$\varepsilon(X) = \begin{cases} \|X^\top X - I\|_F, & X \text{ tall or square,} \\ \|XX^\top - I\|_F, & X \text{ wide.} \end{cases} \quad (7)$$

This metric is applied to `buffer_post`, `g_pre`, and `g_post`. It allows us to distinguish between geometry inherited from the training trajectory before orthogonalization and geometry created directly by the NS map after orthogonalization.

3 Main Results

3.1 Benchmark Results

At fixed depth, the four Muon schedules have nearly the same wall-clock cost. Table 1 shows that all five-step Muon schedules finish in about 1132 seconds. This is an important result: once the number of NS steps and the surrounding optimizer path are fixed, coefficient design barely affect training throughput.

Table 1: Benchmark wall-clock and peak memory for 100M-token runs.

Method	Wall-clock (s)	Peak memory (MiB)
AdamW	1063.99	8380
Stable ($T = 5$)	1133.54	8074
Manual ($T = 5$; $F3_S2$)	1132.34	8074
Fast ($T = 5$)	1132.71	8074
PE ($T = 5$)	1132.58	8074

The main cost difference is the orthogonalization path, not the choice of five-step coefficients. In our implementation, both optimizers are hand-written in PyTorch, so the comparison is not confounded by different kernel maturity. AdamW is faster because it avoids the matrix multiplications and normalization operations required by NS orthogonalization. Within the $T = 5$ Muon family, the coefficients change the scalar polynomial but not the number of polynomial evaluations, so the wall-clock cost stays close. As a result, the practical tradeoff is between additional geometric correction and the amount of validation-loss improvement obtained from it.

3.2 Training Results

Muon consistently improves pretraining quality over the AdamW baseline under the same token budget. Table 2 shows that every Muon variant reaches lower final validation loss and lower validation-loss AUC than AdamW within 100M training tokens. This is the most basic empirical result of the project: introducing semi-orthogonalized matrix updates changes optimization behavior enough to produce a substantial quality gain in this controlled transformer-pretraining setting.

Table 2: Core $T = 5$ training metrics. Mean final loss is reported when multi-seed follow-up is available; otherwise the single-run final loss is shown. AUC is token-normalized, so lower values indicate a better average trajectory.

Method	Mean final loss	Std.	AUC
AdamW	5.3737	–	5.9677
Stable ($T = 5$)	4.5273	–	5.3166
Manual ($T = 5$; $F3_S2$)	4.3176	–	5.0295
Fast ($T = 5$)	4.2892	0.0056	4.9340
PE ($T = 5$)	4.2902	–	4.9338

Fast and Polar Express give the strongest training results in the fixed-depth $T = 5$ comparison. As shown in Figure 1 and Table 2, Fast and PE are numerically the two strongest schedules at fixed depth: Fast achieves the best mean final loss, while PE is extremely close and gives the best AUC. Manual improves clearly over Stable, but it does not match the top two aggressive five-step schedules. The multi-seed follow-up also suggests that this result is reasonably robust: for Fast ($T = 5$), the standard deviation of final loss is about three orders of magnitude smaller than the mean.

Ablation studies on Iteration depth and the PE lower bound To keep this distinction explicit, Tables 3a and 3b summarize these follow-up sweeps separately from the fixed-depth comparison in Table 2.

Within the Manual family, moving from the $T = 5$ hybrid point to the deeper $T = 7$ – 10 range yields a clear one-step improvement in final loss, but the deeper settings then cluster tightly around 4.29. This pattern suggests diminishing returns from additional NS depth once the schedule is already strong enough to reshape the small-singular-value region aggressively. Deeper schedules are modestly slower, which is expected because they execute more NS iterations.

Within the PE family, the lower-bound parameter produces a non-monotonic tradeoff. The strongest final loss appears at $\ell = 3 \times 10^{-3}$, while the best AUC appears at $\ell = 10^{-3}$, and pushing the lower bound down to 3×10^{-5} degrades both metrics. This is consistent with the spectral-map interpretation: changing ℓ changes the interval on which the PE polynomial is designed, but stronger correction at very small singular values is not automatically rewarded by the language-model objective.

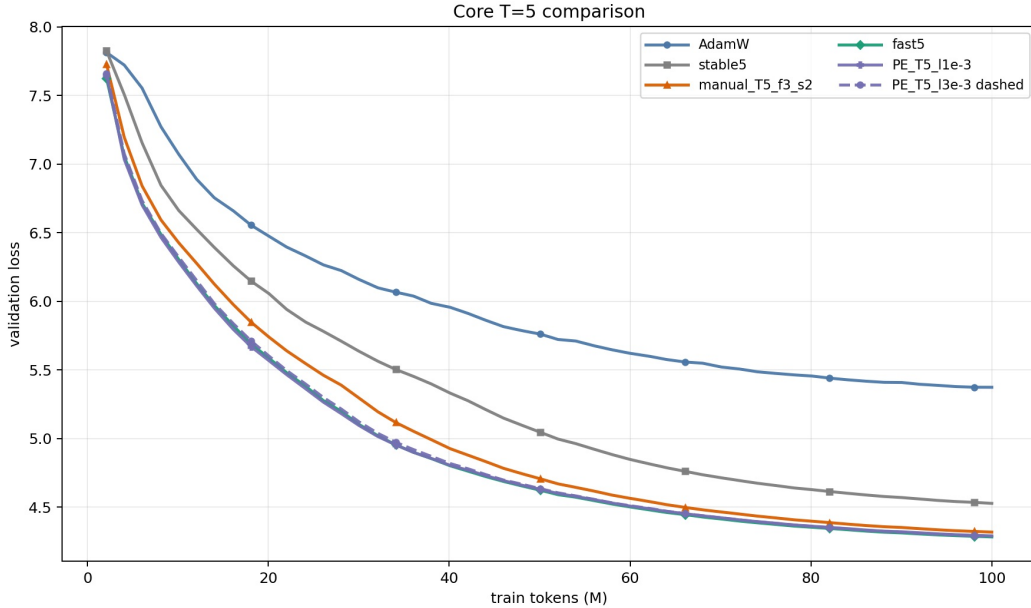


Figure 1: Core $T = 5$ validation-loss curves. The comparison isolates coefficient-sequence effects at fixed depth.

(a) Manual-depth ablation.				(b) PE lower-bound ablation at $T = 5$.			
Schedule	Final loss	AUC	Time (s)	ℓ	Final loss	AUC	Time (s)
$T = 5$ ($F3, S2$)	4.3176	5.0295	1095.57	3×10^{-3}	4.2873	4.9462	1095.88
$T = 7$ ($F4, S3$)	4.2873	4.9312	1107.66	1×10^{-3}	4.2902	4.9338	1094.63
$T = 8$ ($F5, S3$)	4.2890	4.9253	1119.07	3×10^{-4}	4.2998	4.9349	1098.28
$T = 9$ ($F3, S6$)	4.2887	4.9271	1118.57	3×10^{-5}	4.3125	4.9479	1094.97
$T = 9$ ($F4, S5$)	4.2923	4.9268	1118.14				
$T = 9$ ($F5, S4$)	4.2947	4.9286	1118.87				
$T = 9$ ($F9, S0$)	4.2971	4.9308	1120.00				
$T = 10$ ($F5, S5$)	4.2949	4.9282	1131.78				

Table 3: Single-seed ablation studies under the same 100M-token budget. These sweeps probe specific Muon hyperparameters.

3.3 Spectral Geometry Results

Newton–Schulz schedules create visibly different update geometries. Table 4a aggregates the sampled semi-orthogonality errors across training. The first observation is that the error drops from **g.pre** to **g.post** for every schedule, confirming that the NS map is actively doing the intended geometric work at every step. However, the drop varies enormously: the conservative Stable coefficients reduce the error only to 0.9006, which is substantially higher than more aggressive schedules.

The second observation is that the schedules remain clearly separated after orthogonalization, and this separation is monotone in schedule aggressiveness. Stable ends at 0.9006, Fast at 0.6444, Manual $T = 5$ at 0.6866, Manual $T = 9$ at 0.4431, and PE $T = 9$ at 0.3781. This ordering is consistent with the polynomial amplification analysis in Section 3.4: deeper and more aggressive schedules push a larger fraction of the singular-value mass toward 1, and the residual semi-orthogonality error reflects how far the actual update is from the ideal polar factor. Notably, deeper Manual and PE schedules achieve smaller **g.post** error than Fast, even though their final validation losses are comparable or slightly higher—a finding that motivates the separate analysis in Section 3.4.

Schedule sensitivity is larger in attention layers than in MLP layers. Table 4b and the right panel of Figure 2 break the **g.post** error down by module type. The asymmetry is substantial. In attention layers, moving from Fast to PE $T = 9$ reduces the error from 0.5146 to 0.0481—a reduction by a factor of about 10. In MLP layers, the same schedule change reduces the error only from 0.7743 to 0.7081, a change of less than 10%.

This pattern suggests that attention projection matrices and MLP weight matrices experience different spectral regimes during training. MLP matrices appear to maintain a wider spread of singular values that is harder to fully orthogonalize even with deeper schedules, while attention projections respond more cleanly to increased schedule aggressiveness. One

(a) Mean semi-orthogonality errors.

Method	buffer_post	g_pre	g_post
Stable ($T = 5$)	5.0730	5.1962	0.9006
Fast ($T = 5$)	3.4080	3.4584	0.6444
Manual ($T = 5$)	3.2840	3.3425	0.6866
Manual ($T = 9$)	1.9864	2.0471	0.4431
PE ($T = 9$)	2.1216	2.1781	0.3781

(b) Mean **g_post** error by module.

Method	Attention	MLP
Fast ($T = 5$)	0.5146	0.7743
Manual ($T = 9$)	0.1776	0.7086
PE ($T = 9$)	0.0481	0.7081

Table 4: Spectral summaries for overall and module-specific geometry. Lower semi-orthogonality error means the update is closer to the polar-factor target.

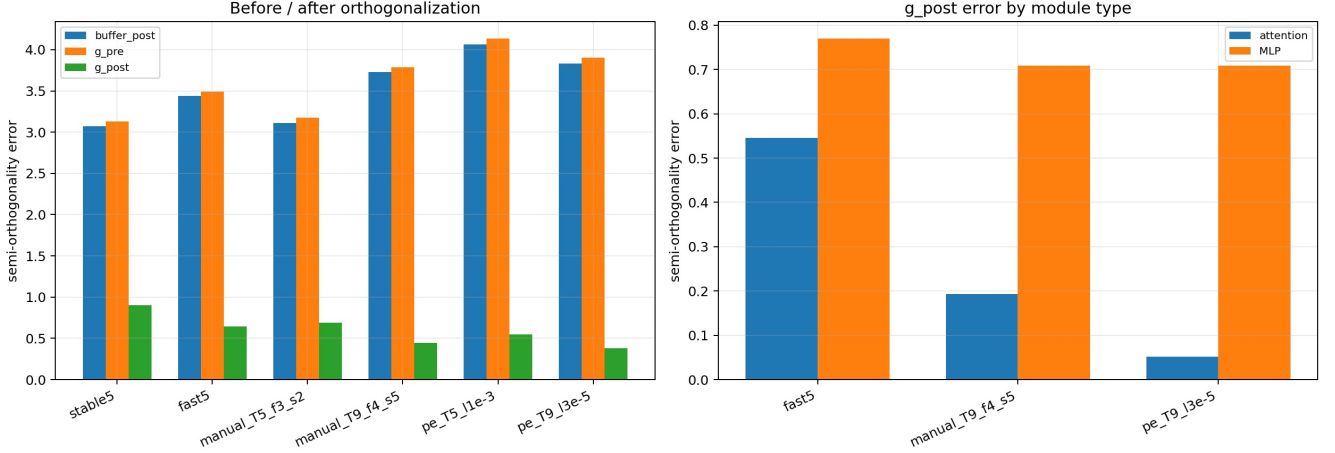


Figure 2: Spectral diagnostics from sampled update matrices. Left: error before and after the NS map. Right: module-level **g_post** error for representative schedules.

interpretation is that the effective rank of gradient directions is lower in attention layers, so a stronger polynomial can more efficiently collapse the singular-value distribution toward 1. The result has a practical implication: if the goal is to tighten update geometry in a specific module, schedule depth matters most for attention layers, and adding NS steps beyond $T = 5$ yields diminishing returns in MLP layers specifically.

3.4 Spectral Map Analysis

The geometry differences observed above have a precise mathematical origin: different schedules define different composed scalar maps on singular values, and these maps determine how aggressively small singular directions are promoted toward the polar-factor target.

Single-step behavior For a schedule with T NS steps, the composed scalar map p_T is obtained by iterating $p_k(\sigma) = a_k\sigma + b_k\sigma^3 + c_k\sigma^5$. Figure 3 shows the single-step behavior. The Fast coefficient map amplifies small singular values much more strongly than the Stable NS map, but its derivative passes through zero and becomes negative in part of the input range. This means Fast behaves less like a conservative contraction and more like an aggressive spectral reshaping rule: it is not monotone in singular-value rank, and the mapping is order-reversing for certain input values. The key distinction between the Fast and Stable schedules is that instead of strictly attracting all singular values in $[0, 1]$ toward 1, the Fast map pushes them toward the range $[0.7, 1.3]$ and tolerates some oscillation around the target. The fact that Fast achieves the best training outcome suggests that validation loss is not monotone in the degree of semi-orthogonality: a moderate level of orthogonalization is beneficial, but maximally tight orthogonality is not necessary.

Composed maps and amplification After several steps, the local single-step differences accumulate into very different overall maps. Left panel of Figure 4 illustrates this: Stable barely lifts small singular values, Fast produces a noticeably steeper rise, and deeper Manual or PE schedules push small singular values much closer to 1. The right panel illustrates the composed-map behavior of PE under different lower-bound choices. The convergence behavior depends strongly on the lower-bound assumption ℓ : PE with $\ell = 10^{-3}$ effectively attracts all singular values above 10^{-3} to near 1, while the alternative with $\ell = 3 \times 10^{-5}$ exhibits oscillation across a wider input range, failing to produce a clean fixed-point attraction.

Table 5 quantifies the amplification at the representative input $\sigma = 10^{-5}$. The contrast is striking: Stable has an amplification ratio of only 32, leaving a singular value of 10^{-5} still negligibly small at 3.2×10^{-4} . Fast raises the

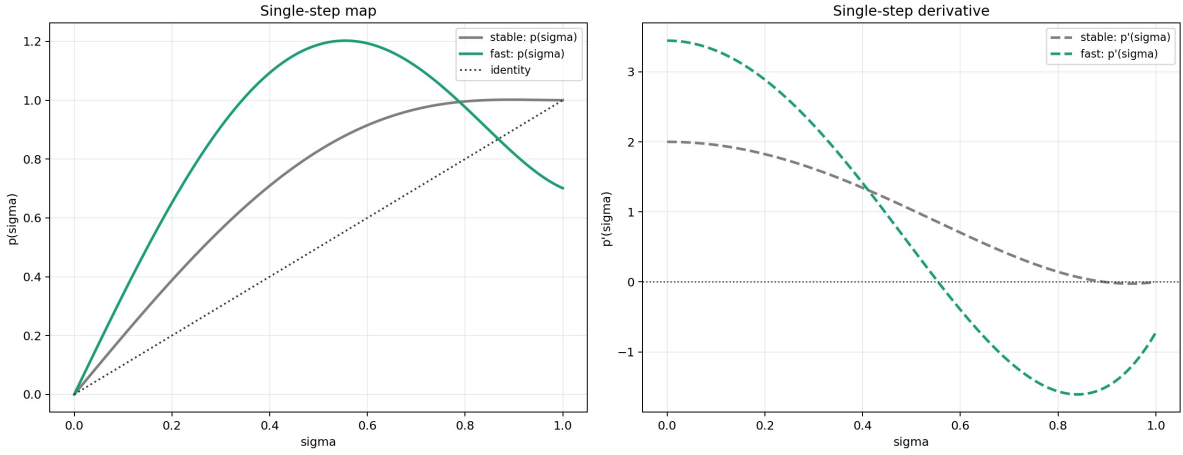


Figure 3: Single-step Newton–Schulz maps and derivatives for the Stable and Fast coefficient choices. The derivative panel shows where the map is locally amplifying or reversing order in singular-value space.

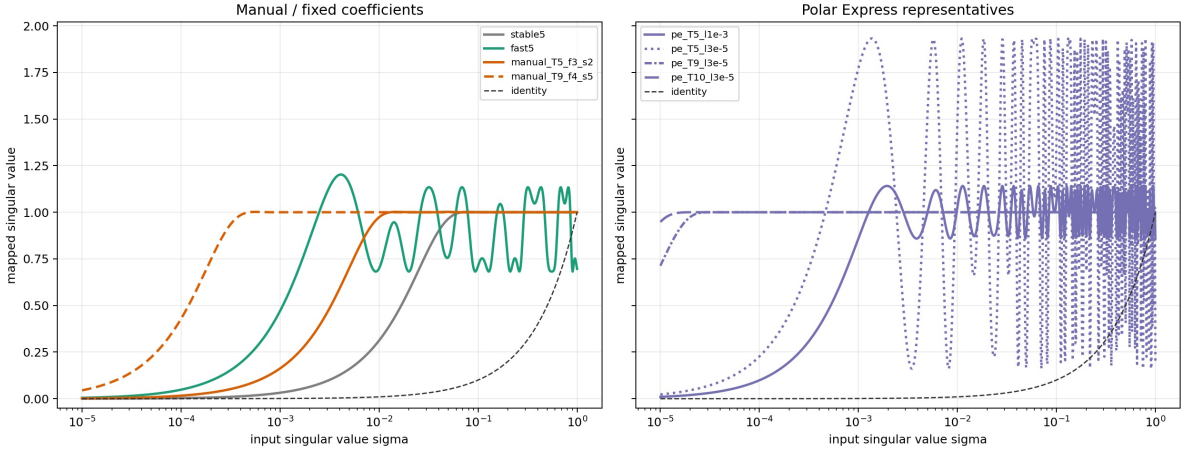


Figure 4: Composed singular-value maps $p_T(\sigma)$. The dashed identity line is the input; colored curves show the output after a full schedule. Schedules closer to the constant-1 line are closer to the ideal polar factor.

amplification ratio to 485, but the output is still far from the polar target. Manual at $T = 9$ reaches an amplification ratio of 4502 and pushes the output to 4.5×10^{-2} , while PE at $T = 9$ achieves an amplification ratio of 7.1×10^4 , mapping 10^{-5} all the way to 0.713—within striking distance of the ideal value of 1. These numbers make the geometric improvement in Table 4a interpretable: it is not that different schedules apply more iterations to the same map, but that they define fundamentally different maps, and a single application of a well-designed map can accomplish what many steps of a conservative map cannot.

Table 5: Small-singular-value amplification implied by the composed maps.

Method	$p_T(10^{-5})$	$p_T(10^{-5})/10^{-5}$
Stable	3.20×10^{-4}	32
Fast	4.85×10^{-3}	485
Manual ($T = 9$)	4.50×10^{-2}	4502
PE ($T = 9$)	7.13×10^{-1}	71309

Spectral perturbation structure Figure 5 plots the derivative $p'_T(\sigma)$ of each composed map. The most striking feature is that *all* schedules saturate at the ± 20 clip limit for $\sigma \lesssim 10^{-3}$: in the deep spectral tail every schedule is equally extreme, so that region alone cannot explain the performance differences observed between schedules.

The informative region is the intermediate singular-value range $\sigma \in [0.01, 0.1]$, where gradient mass in typical language-model training is likely to concentrate. Here the schedules diverge qualitatively. Stable shows a smooth, monotonically decreasing derivative with no sign changes, meaning it preserves the relative ordering of singular directions throughout.

Fast, by contrast, exhibits large oscillations with sign changes across this range, indicating that it actively inverts the importance ranking of some singular directions during orthogonalization. The deeper Manual and PE schedules have relatively flat derivatives in the moderate- σ range, consistent with their role as maps that collapse the dominant spectral mass toward 1 cleanly. PE with $\ell = 3 \times 10^{-5}$ is an exception: the oscillatory behavior shifts to smaller σ , suggesting its polynomial is being stretched over a spectral interval that covers directions likely dominated by noise.

These observations do not yield a simple causal explanation for the non-monotone relationship between orthogonality and training loss. Rather, they suggest that the schedules differ not only in *how much* they reshape the spectrum but in *which part* of the spectrum they perturb most and in what manner. Whether a given perturbation aids or hurts training ultimately depends on how the affected singular directions align with the loss landscape—a quantity not directly readable from the map structure.

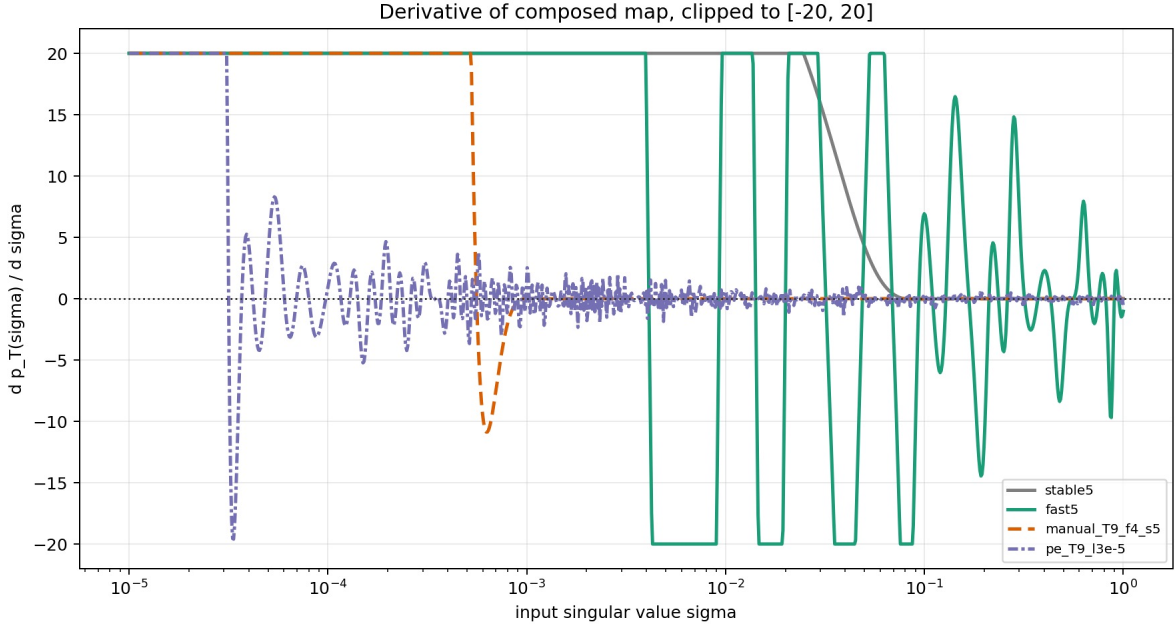


Figure 5: Derivatives of the composed singular-value maps, clipped to $[-20, 20]$. The schedules are indistinguishable in the deep spectral tail ($\sigma \lesssim 10^{-3}$) where all saturate; the qualitative differences emerge in the intermediate range where gradient mass is concentrated.

Together, the composed-map analysis reveals that schedule choice shapes not only how aggressively the NS iteration pushes singular directions toward the polar target, but also which part of the gradient spectrum is most actively perturbed. Schedules that achieve better semi-orthogonality do so by applying stronger, qualitatively different maps—yet this does not translate monotonically into lower validation loss. The derivative structure suggests that each schedule perturbs a different spectral region in a different manner, but whether any particular perturbation is beneficial or harmful depends on how those singular directions align with the loss landscape, which the map alone cannot determine.

4 Conclusion

This project asked how different Newton–Schulz coefficient schedules change the spectral geometry of Muon updates, and how those geometric changes relate to transformer pretraining behavior under a fixed experimental stack. The main conclusion is that the schedule matters because it changes the geometry of the actual matrix update, not merely because it changes an implementation detail inside the optimizer. At fixed depth, fast Muon and well-chosen PE schedules are much stronger than the stable NS schedule. This indicates that aggressive small-singular-value amplification is useful in the early-scale pretraining regime studied here.

The spectral evidence gives a more detailed picture. Different schedules produce different composed singular-value maps, and those maps are reflected in the measured geometry of real update matrices. Deeper Manual and PE schedules substantially reduce post-orthogonalization error, especially in attention layers. However, this cleaner geometry does not translate monotonically into lower validation loss: in the representative multi-seed runs, Fast remains the strongest practical baseline at 100M tokens, while the more orthogonal $T = 9$ schedules are close but not better.

Overall, the experiments suggest a balanced view of NS schedule design in Muon. Fast coefficients are highly effective for the present small-transformer pretraining setting, while Manual and PE schedules provide a more controllable way to reshape the spectrum and study the geometry of the update. The most interesting open question left by the experiments

is not simply how to make the update more orthogonal, but how the composed singular-value map should be designed to perturb the *right* spectral directions—those that are informative for the loss—rather than merely maximizing amplification across the board.

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